MODEL OF INHERITANCE

Various hereditary characteristics are controlled by two genes that occur in pairs - one gene from each of the mother and father.

One gene in a pair may ***dominate*** the other, meaning that it "masks" the presence of the ***recessive*** gene. Hence, the characteristic will show according to the dominant gene.

Capital letters represent the dominant gene; lower case letters represent the recessive gene.

Example

Gregor Mendel did experiments on pea plants. He designated "T" for the dominant tall trait, while "t" represented the recessive dwarf trait.

TT and Tt represent the dominant tall trait.
tt represents the recessive dwarf trait.

Definitions

Allele - one of the given forms of a gene.

e.g. T - tall, t - dwarf

Homozygous (pure-bred) gene - made up of two identical genes for a particular characteristic.

e.g. TT, tt

Heterozygous (hybrid) gene - the paired genes for a particular characteristic are different.

e.g. Tt

Genotype - the genetic make-up of an organism (the actual genes).

e.g. TT, Tt. tt

Phenotype - the appearance of an organism as determined by the genotype

e.g. TT - tall Tt - tall tt - dwarf

Example 1

A pure-bred (homozygous) tall pea plant is crossed with a pure-bred (homozygous) dwarf plant. determine the genotypes and phenotypes of the first (F_1) and second (F_2) generation offspring.

TT x tt
♂ ♀

♀ \ ♂	T	T
t	Tt	Tt
t	Tt	Tt

F_1 genotypes Tt Tt Tt Tt

phenotypes 100% tall

Tt x Tt
♂ ♀

♀ \ ♂	T	t
T	TT	Tt
t	Tt	tt

F_2 genotypes $\underbrace{\text{TT Tt Tt}}$ $\underbrace{\text{tt}}$

phenotypes 75% tall 25% dwarf

Note

A ***Punnett square*** is used to work out the combination of alleles for each cross.

Example 2

In a particular species of fish, black body colour (B) dominates red body colour (b). Predict the ratio or percentage of black fish to red fish if a heterozygous black fish is crossed with a homozygous red fish.

Bb x bb
♂ ♀

♀ \ ♂	B	b
b	Bb	bb
b	Bb	bb

genotypes $\underbrace{\text{Bb Bb}}$ $\underbrace{\text{bb bb}}$

phenotypes 50% black 50% red

$\therefore$ <u>Ratio of black to red is 1:1</u>